Department
of
Mechanical Engineering

LAB MANUAL

Mechanical Engineering

Production Laboratory-IV (ME)

ME-423-B

Final year/ 7th Semester

Program Outcomes:

PO1 **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6 **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8 **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9 **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10 **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12 **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Vision of the Institution

"To be a leading educational institute that provides quality technical education and conducts research to produce knowledge-rich professionals for meeting the dynamic needs of the industry and society."

Mission of the Institution

1. To impart quality technical education to the students to make them globally competent engineers, contributing to the development of the nation and world at large.
2. To imbibe ethical values, scientific and industrial temperament, and spirit of innovation among students.

Vision of the Department

To provide quality technocrat compatible with global standard in the field of mechanical engineering who can contribute to society through innovations, entrepreneurship & sustainable development.

Mission of the Department

M1. To impart highest quality in fundamentals, technical knowledge & scientific education to the learners to make them globally competitive mechanical engineers.

M2. To promote liaison with world class industries & educational institutions for excellence in teaching, research & consultancy practices.

M3. To make students, life-long learners, ethically and technically capable to build their careers in terms of professions and personality.

Program Specific Outcomes:

PSO1 To empower the students to understand the basic concepts of mechanical engineering. In addition to this, application of theoretical and practical skills to solve complex mechanical engineering problems with the use of advanced software tools to arrive at appropriate and cost-effective solutions.

PSO2 To develop entrepreneurship skills compatible with industrial readiness and can pursue higher studies in Mechanical Engineering or interdisciplinary programs with the concern for sustainable development of society & high regard of ethical values.

TYPE OF COURSE B.E. Mechanical (ME- 423 B)

NAME OF COURSE: - Production Lab – IV

METAL CUTTING LAB	
1.	To study and calibrate a 2D lathe dynamometer
2.	To plot tangential and feed force as function of cutting speed and feed in turning, also calculate and plot cutting power as a function of feed rate
3.	To study the effect of rake angle land chip thickness ratio on shear angle in orthogonal machining

MACHINE TOOL LAB	
1.	Explain term machinability and conduct cuttability test on a drilling machine for comparing machinability of different metals
2.	To study the tool makers microscope and checking the alignments of screw thread and cutting tool (five elements of workpiece are checked, outer dia, root dia, pitch, depth, thread angel) (measuring angles end cutting edge angle, side cutting edge angle, lip angle, top back rake angle, clearance angle)
3.	To calibrate the Pneumatic comparator and measure angles and tapper of given test materials
4.	Study of universal tool cutter grinder

COURSE OUTCOMES: -

1. To develop the knowledge of application and analysis of metal cutting (orthogonal cutting)

2. To Evaluate and analyze the effect of metal cutting parameter on the process.

3. To identify the influence of different factors on machinability.

4. To study the machine tools and precision measurement.

Metal Cutting Lab
Experiment – 1

Object: - To study and calibrate 2-D lathe Dynamometer.

Apparatus: -

1.) A 2-D lathe dynamometer.
2.) A centre lathe.
3.) Weight.
4.) Lathe.

Procedure: -

1. Make a dimensional sketch of dynamometer.
2. Sketch the circuit of wheat store bridge for the feed (longitudinal) and tangential (vertical) components of circuit force.
3. Explain in your own words the principle of working of dynamometer and calculate the theoretical sensitivity in micrometer per kg. of applied load.
4. Statically load the dynamometer for

(a) Vertical force

(b) Horizontal force in steps

and record the bridge readings. After reaching the maximum load unload in gradual steps and record the readings.

5. Plot the calibration graph, find the actual sensitivity and compare it with the theoretically calculated value.
6. Plot the calibration graph, find the actual sensitivity.

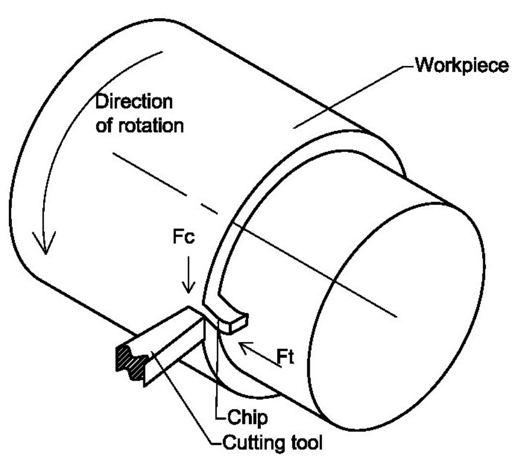


Figure 1: Lathe turning operation

Observations:

(i) Dynamometer dimensions
(ii) Calibration readings

Least count $\frac{0.1}{50}$ = 0.002 mV

Net value = dial reading × Least count

Range selector = 0.1 mV

Average net value = $\frac{\text{Net value on loadoing+Net value on unloading}}{2}$

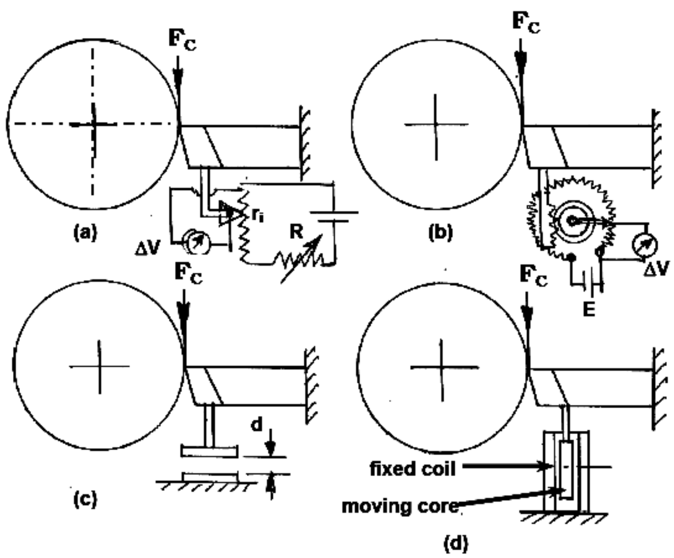


Figure 2: Electrical transducers working based on deflection measurement
(a) linear pot (b) circular pot (c) capacitive pick up (d) LVDT type

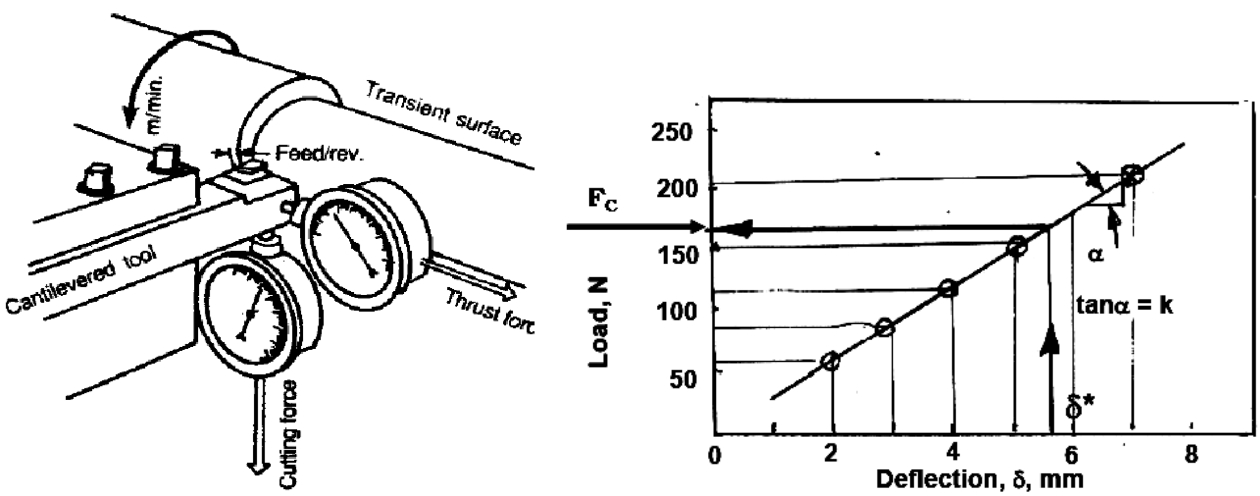


Figure 3 deflections versus load

Theory :

The dynamometer function on the wheat stone bridge principle of balancing a circuit having four strain gauges resistance. The dynamometer measured the tangential and feed component of cutting force which act on vertical and horizontal plane respectively. To sense each component of the cutting force four strain gauges are mounted on elastic member of the dynamometer.

Majority of dynamometer used for measuring the tool forces use the defections of strains caused in the components, supporting the tool in metal cutting.

Governing Equation Formulation: This work is based on the measurement of elastic deflection of the cutting tool subjected to a cutting force. The cutting tool can be approximated to a cantilever which is held by the tool holder and in turn the dynamometer which is fixed. Figure 3 shows the graphical effect of the cutting force of a rotating work-piece on the cutting tool (which is modeled as a cantilever). The dotted cutting tool sketch shows the deflection effect Δ from the cutting force Fc.

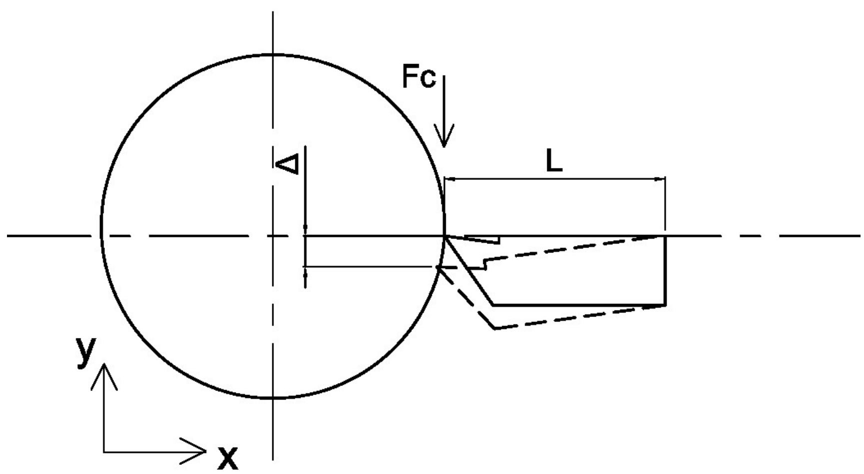


Figure 4: Deflection effect of the cutting tool from cutting force, adapted from

The cutting tool being subjected by a cutting force F_c causes an elastic deflection Δ which is linearly proportional to the force. The equation of the deflection curve of a cantilever is

$$\upsilon = \frac{W(3Lz^2 - z^3)}{6EI} \tag{1}$$

The deflection at the tip of the cutting tool which is , W = F_c and by setting z = L in equation (1),

$$\Delta = \frac{FcL^3}{3EI} \tag{2}$$

For a given cutting tool and the holder, E and I are constant, L is also constant for a particular length; therefore, Δ is linearly proportional to F_c and equation (2) can be written as

$$\Delta = \alpha F_c, \tag{3}$$

where α is a constant of proportionality

During calibration of a two component lathe dynamometer, there may be cross-sensitivity between the two force components; hence the equations after Boothroyd are used in measuring the cutting forces given as:

$F_c = \frac{A}{X} R_c - \frac{C}{X} R_t$ (4)

$F_t = \frac{D}{X} R_t - \frac{B}{X} R_c$ (5)

Where X=AD – BC

Dynamometer are made in deferent grads of accuracies according to purpose

Grade	accuracy
Workshop or regular	± 5.07s.
Calibration	± 1.09.
Research	± 0.59.

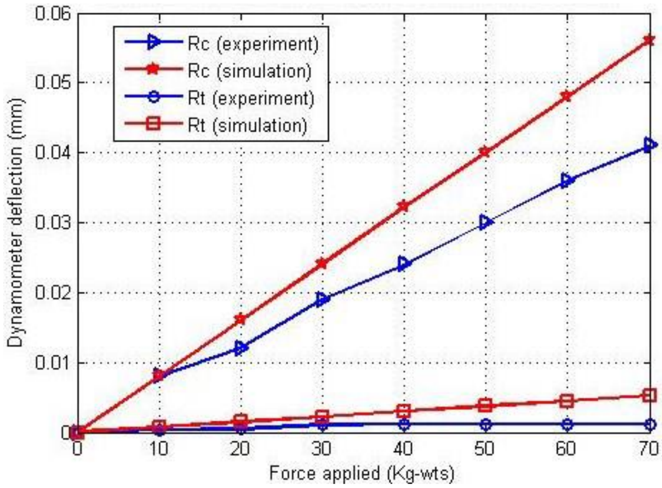


Fig 5: Dynamometer calibration curve; loading the vertical component (cutting force, F_c)

Observation Table:-

S.No.	Applied load (k.g.)	Bidge reading loading (mV)	Unloading (mV)

Total load = kg

Total (mV) = …………….

Calculation :

Sensitivity = $\frac{Total\,(mV)}{Total\,load}$

=

= ………………. mV/kg

Calibration:-

y = mx + c

m = $\left(\frac{y2-y1}{x2-x1}\right)$ = $\frac{20-10}{10-5}$ = 2

c = 0

So 1 mV = 2kg (load)

Result :

Operating Instruction for bridge balance unit :-

1. Place the sensing unit of dynamometer in proper position.
2. With the help of cables, carefully connect PU1socket on sensing unit ot PU1 socket on strain gauge amplifier unit Repeat the same PU2 channel.
3. Connect the instrument to 230 V, 1 phase supply.
4. Ensure the range select switch is in full clock – wire position channel select switch in mid position.
5. By putting the socket 2 in Batt, check position, see if batteries are in good condition. If the meter pointer doesn't snow a reading btw. 4 to 4.5 replace the batteries. For battery replacement backside cover is required to be removed. On the back – cover you will find a battery housing switch accommodated 6 No. of 1.5 V dry cell of standard variety.
6. Now keep the switch 3 in read position and take the range selector switch to the most sensitive position and see if the micro voltmeter (m3) indicates zero. If m1 is not indicating zero (w.r.t. user) adjust a small screw. Adjust on left hand side of the instrument. This adjustment must be done very carefully as 22 turn trimming pot whci makes this zero. Adjustment is very delicate. Again take the switch 4 range selector switch to its least sensitive range.
7. Now the instrument is ready for use. Switch on the AC mains by putting switch 1 in down word position.
8. Take sw 5 to left hand position and observe that bridge excitation voltage of 5 Volts is indicated on the M2 meter. When sw 5 volt is taken to right hand position, bridge exciation voltage of 5 Volts for PU2 channel will be indicated by m2 meter.
9. Obtain balance on 300 microvolt range by operating balance plots the micro voltmeter for both the channels, under specified conditions. By referring to the calibration curves provided, you can come to know about magnitude of force measured with both channels.

Experiment – 2

Object:- To plot tangential and feed forces V/s cutting speed and feed during the turning operation while cutting dry.

Apparatus required: Centre lathe, 2-D lathe tool dynamometer, mild – steel work-piece.

Theory :

Force relationship in orthogonal cutting:

There are no of forces acting on chip during metal cutting. Relationship among these forces were established by. Merchant's theory with following assumptions -

1. Cutting Velocity always remains constant.
2. Cutting edge of tool remains sharp through cutting.
3. There is no contact between Work-piece & tool flank.
4. There is no sideways flow or chip.
5. Only continuous chip is produced.
6. No. built up edges.
7. Inertia forces of chip neglected.
8. Behavior of chip is like that of a free body which is in state or stable equilibrium due to action of resultant forces which are equal, opposite & collinear.

$F = F_c \sin \propto + F_t \cos \propto$

$N = F_c \cos \propto + F^t \sin \propto$

$F_s = F_c \cos \varphi - F_t \sin \varphi$

$F_n = F_t \cos \varphi + F_c \sin \varphi$ $\quad\quad \frac{F}{N} = \tan\beta = \mu$

here β = angle or friction

$\propto$ = Rake angle

$\tan \varphi = \frac{x \cos \propto}{1 - x \sin\propto}$

Force on single point tool in turning in case of orthogonal cutting. Only 2 components of forces.

$R = \sqrt{Fc^2 + Ft^2}$

Procedure :-

- Read carefully operating instructions for bridge balance unit & operate instrument accordingly. Start machine& engage tool for give cutting, take speed at different conditions.
- Given carefully cuts by varying spindle R.P.M. keeping other parameters constant. Start from minimum R.P.M. and increase the speed to given max^m.
- For variation of feed, take increasing value of feed in steps keeping other parameter constant plot variation of cutting forces.

1. **Cutting force for varying cutting speed:**

depth of cut = 0.5 m.m.
feed = 0.325 mm/rev
work dia = 56 mm.

rpm. = Select five suitable steps one in higher side & fo8ur in lover side of the suitable r.p.m. for the given dia. of w/p .

2. **For varying feed:**

Depth of cut = 0.5 mm.
r.p.m = 106 (const)
work dia = 56 m.m.
Feed = 0.0036, 0.0051, 0.0072, 0.0102, 0.014, 0.0128

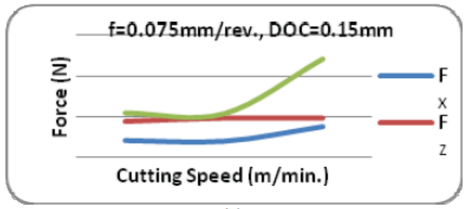
Observation table :

(1.) For varying cutting speed.

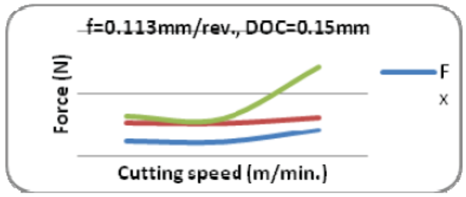
S.No.	cutting speed		tangential force (PU_1)		feed force (PU_2)	
	R.P.M.	m/min	dial reading	force (N)	dial reading	force (N)

(2) **For varying feed.**

S.No.	feed/rev.	r.p.m.	tangential force (PU_1)		feed force (PU_2)	
			dial reading	force (N)	dial reading	force (N)



(a)



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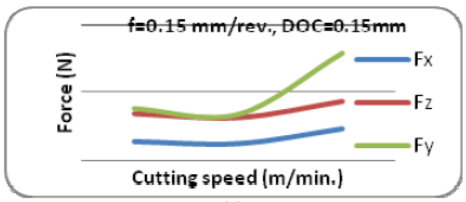


(c)

Fig.6. Variation in forces with cutting speed for d=0.15mm (a) f=0.075mm/rev (b) f=0.113 mm/rev. (c) f=0.15 mm/rev.

Conclusions: Depth of cut was found to be the most influential parameter affecting the three cutting forces followed by the feed. Cutting speed was least significant in case of axial and radial force models but was not significant for the regression model of cutting force. For the surface roughness predictions, the model developed from the analysis was found insignificant.

<u>Tool geometry:</u>

- Front clearance angle = 6°
- Side clearance angle = 6°
- Side rake angle = 6°
- Back rake angle = 10°
- And cutting edge angle = 6°
- Nose radius = 20.5 mm.

<u>Calculation:</u>

(1.) For varying cutting speed:

(i) *Cutting speed (m/min):* D = 56 mm. = 0.056 m.

$S_1 = \barwedge DN_1 = \barwedge D \times \quad =$

$S_2 = \barwedge DN_2 = \barwedge D \times \quad =$

$S_3 = \barwedge DN_3 = \barwedge D \times \quad =$

$S_4 = \barwedge DN_4 = \barwedge D \times \quad =$

$S_5 = \pi DN_5 = \pi D \times \qquad =$

(ii) Tangential force:

The reading of calibration curve

$1\ \mu v = 2$ k.g.

$f_1 = 0.4 \times 50 \times 100$

$= 2000$ N $\qquad \begin{bmatrix} 1\ \text{k.g.} = \ ^1/_2\ \mu v \\ 1\text{N} = 5\ \mu v \end{bmatrix}$

$F_1 = 0.34 \times 5 \times 1000 = 1700$ N

$F_3 = 0.30 \times 5 \times 1000 = 1500$ N

$F_4 = 0.32 \times 5 \times 1000 = 1500$ N

$F_5 = 0.42 \times 5 \times 1000 = 2100$ N

(iii) feed force:

$F_1 = 0.1 \times 5 \times 1000 = 500$ N

$F_2 = 0.08 \times 5 \times 1000 = 400$ N

$F_3 = 0.06 \times 5 \times 1000 = 300$ N

$F_4 = 0.08 \times 5 \times 1000 = 400$ N

$F_5 = 0.14 \times 5 \times 1000 = 700$ N

(2.) For varying feed rate:

(i) *tangential force from calibration curve* $\begin{cases} 2\ \text{k.g.} = 1\ \mu v \\ 1\ \mu v = 1\ \text{N} \end{cases}$

tangential force = μv × load/μv

$F_1 = 0.12 \times 5000 = 600$ N

$F_2 = 0.14 \times 5000 = 700$ N

$F_3 = 0.16 \times 5000 = 800$ N

$F_4 = 0.20 \times 5000 = 1000$ N

$F_5 = 0.26 \times 5000 = 1300$ N

$F_6 = 0.30 \times 5000 = 1500$ N

(ii) *feed force from calibration curve* $\begin{cases} 2\ \text{k.g.} = 1\ \mu v \\ 5\ \mu v = 1\ \text{N} \end{cases}$

feed force = μv × load/ μv

$F_1 = 0.04 \times 5 \times 1000 = 200$ N

$F_2 = 0.05 \times 5 \times 1000 = 250$ N

$F_3 = 0.044 \times 5 \times 1000 = 220$ N

$F_4 = 0.06 \times 5 \times 1000 = 300$ N

$F_5 = 0.08 \times 5 \times 1000 = 400$ N

$F_6 = 0.09 \times 5 \times 1000 = 450$ N

Result:

Experiment – 3

Object: To study the effect of rake angle on the chip thickness ratio and the shear angle in orthogonal cutting.

Apparatus: lathe, single point turning tools with varying side rake angle profile projection and weak.

Theory:

Metal cutting & single point cutting tool: Metal are desired forms through various process like removing the unwanted materials from of chips through machining.

Basic elements of machining:

(i) workpiece
(ii) tool
(iii) chip

type of cutting:

(i) orthogonal cutting
(ii) oblique cutting'

Geometry of single point cutting tool: Tool signature is the process of standardization specityiing principle tool angles of a single point cutting tool:-

a) Shank
b) Face
c) Flank
d) Base
e) Heel
f) Nose radius

Shear angle relationship:

Helpful to predict position of shear plane (angle ϕ)

• **Relationship between-**

- ✓ Shear Plane Angle (ϕ)
- ✓ Rake Angle (α)
- ✓ Friction Angle(β)

Several Theories

Earnst-Merchant (Minimum Energy Criterion):
Shear plane is located where least energy is required for shear.

Assumptions:-

• Orthogonal Cutting.

• Shear strength of Metal along shear plane is not affected by Normal stress.

• Continuous chip without BUE.

• Neglect energy of chip separation.

Assuming No Strain hardening:

Condition for minimum energy, $\frac{dF_c}{d\phi} = 0$

$$\frac{dF_c}{d\phi} = \tau\, t_u\, b \operatorname{Cos}(\beta - \alpha)\left[\frac{cos\emptyset cos(\phi+\beta-\alpha) - sin\emptyset sin(\phi+\beta-\alpha)}{sin^2\phi\, cos^2(\phi+\beta-\alpha)}\right]$$

$\therefore \cos(\phi)\cos(\phi + \beta - \alpha) - \sin\phi\sin(\phi + \beta - \alpha) = 0$

$\cos(2\phi + \beta - \alpha) = 0$

$2\phi + \beta - \alpha = \frac{\pi}{2}$

$$\boxed{\therefore\ \phi = \frac{\pi}{2} - \frac{1}{2}(\beta - \alpha)}$$

<u>**Tool signature (Angles):**</u>

1.) Rake angle:

(i) *Top rake:* Angle btw. the face & the plane parallel to the base. It is known as top rake. It inclined towards the shank.

(ii) *Side rake:* When measured towards the side of the tool.

2.) **Lip angle:** It is known as angle of kneeness. It is angle btw. the face & shank.

3.) **Clearance angle:** the angle formed by the front or side of the tool which are adjacent & below the cutting edge clearance angle can be:

i.) Front clearance angle: Angle just below point helpful to present lubing.

ii.) Side clearance: Direct the cutting through to the metal area adjacent to cutting edge.

4.) **Relief angle:** Angle btw. the of tool & perpendicular line (imaginary) drawn from the cutting point to the base of the tool.

5.) **Cutting angle:** Total cutting angle in the angle formed btw. the tool face & time through the point which is tangent to the machined surface to the work at that point.

Formula: In case of orthogonal cutting the shear angle is given by:

$$\operatorname{Tan}\varphi = \frac{r \operatorname{Cos}\propto}{1 - r \operatorname{Sin}\propto}$$

Where: $\propto$ = rake angle

r = chip thickness ratio = $\frac{t}{t_c}$

Chip thickness can be found out by the

$t_c = \frac{m_c}{l_c \times w_c \times s}$

where: m_c = mass of the chip

l_c = length of the chip

w_c = width

s = density (7.83 gm/cm^3)

Observation:

Pipe outside diameter of the ($\Delta_\circ$) = 62 m.m.

Pipe inside diameter (Δ_i) = 55 m.m.

Mean dia. $\left(\frac{\Delta_i + \Delta_\circ}{2}\right) = \frac{62+55}{2}$ = 58.5 m.m.

r.p.m. (N) = 236

Cutting velocity ($\barwedge$ DN) = $\frac{\barwedge \times 58.6 \times 236}{1000}$ = 43.37 $^{m}/_{min}$

Feed = 0.0014 $^{in}/_{rev.}$

- dendity of material (chip) = 7.83 gm/c.c
- thickness of uncut chip = feed of lathe.

So t = 0.005 $^{in}/_{rev.}$

f = 0.127 mm/rev

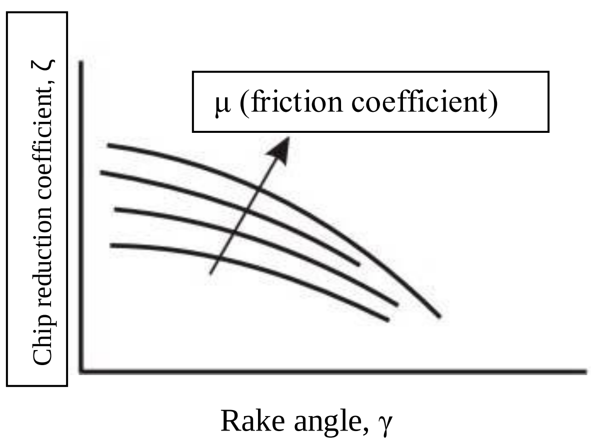


Fig. 3 *Role of rake angle and friction on chip reduction coefficient*

Observation table:

S. No.	**Rake angle ∝ (degree)**	**Length of chip (m.m.)**	**Width of chip (m.m.)**	**Mass of chip (g.m.)**	**Thickness of chip (t_c)**	**Avg. (t_c)**

Calculation:

Thickness of chip (t_c) = $\frac{m_C}{t_c \times w_c \times s}$

(i) for rakes angle ($\propto$) = 4°

$t_{c1} = \frac{0.0878 \times 1000}{5.7 \times 3.3 \times 7.83}$ = 0.596 m.m.

$t_{c2} = \frac{0.0781 \times 1000}{5.1 \times 3.5 \times 7.83}$ = 0.558 m.m.

$t_{c3} = \frac{0.0759 \times 1000}{5.2 \times 9.4 \times 7.83}$ = 0.548 m.m.

avg. thickness = 0.567 m.m.

(ii) for rakes angle ($\propto$) = 8°

$t_{c1} = \frac{0.0937 \times 10^3}{9.4 \times 3.2 \times 7.83}$ = 0.0.397 m.m.

$t_{c2} = \frac{0.0691 \times 10^3}{7.1 \times 3.0 \times 7.83}$ = 0.414 m.m.

$t_{c3} = \frac{0.0620 \times 10^3}{5.9 \times 3.1 \times 7.83}$ = 0.437 m.m.

avg. thickness = 0.416 m.m.

(iii) for rakes angle ($\propto$) = 4°

$t_{c1} = \frac{0.0762 \times 10^3}{7.2 \times 3.3 \times 7.83}$ = 0.409 m.m.

$t_{c2} = \frac{0.0876 \times 10^3}{8.5 \times 3.2 \times 7.83}$ = 0.411 m.m.

$t_{c3} = \frac{0.0559 \times 10^3}{6.9 \times 3.5 \times 7.83}$ = 0.295 m.m.

avg. thickness = 0.371 m.m.

(iv) for rakes angle ($\propto$) = 16°

$t_{c1} = \frac{0.0575 \times 10^3}{7.9 \times 3.5 \times 7.83}$ = 0.265 m.m.

$t_{c2} = \frac{0.0531 \times 10^3}{8.2 \times 3.4 \times 7.83}$ = 0.243 m.m.

$t_{c3} = \frac{0.0526 \times 10^3}{8.3 \times 3.6 \times 7.83} = 0.224$ m.m.

avg. thickness = 0.244 m.m.

(v) for rakes angle ($\propto$) = 20°

$t_{c1} = \frac{0.0463 \times 10^3}{7.8 \times 2.8 \times 7.83} = 0.261$ m.m.

$t_{c2} = \frac{0.0444 \times 10^3}{7.6 \times 3.6 \times 7.83} = 0.207$ m.m.

$t_{c3} = \frac{0.0377 \times 10^3}{7.5 \times 3.4 \times 7.83} = 0.188$ m.m.

avg. thickness = 0.218 m.m.

<u>chip thickness ratio & shear angle:</u>

(i) For rake angle ($\propto$) = 4°

t_c = 0.567 mm, t = 0.127 m.m.

chip thickness ratio (x_1) $= \frac{t}{t_c} = \frac{0.127}{0.507} = 0.223$

shear angle $\varphi_1 = \tan^{-1}\left(\frac{x\ Cos2}{1-x\ sin2}\right)$

= 12.73°

(ii) For rake angle ($\propto$) = 8°

t_c = 0.416 mm, t = 0.127 m.m.

chip thickness ratio (x_2) $= \frac{0.127}{0.567} = 0.305$

shear angle $\varphi_2 = \tan^{-1}\left(\frac{0.305 \times Cos\ 8}{1-0.305 \times Sin\ 8}\right)$

= 12.73°

(iii) For rake angle ($\propto$) = 12°

t_c = 0.371 mm, t = 0.127 m.m.

$x_3 = \frac{0.127}{0.371} = 0.342$

$\varphi = \tan^{-1}\left(\frac{0.342 \times Cos\ 12}{1-0.342 \times Sin\ 12}\right)$

= 19.8°

(iv) For rake angle ($\propto$) = 16°

t_c = 0.244 mm, t = 0.127 m.m.

$x_4 = \frac{0.127}{0.244} = 0.520$

$\varphi = \tan^{-1}\left(\frac{0.520 \times Cos\ 16}{1-0.520 \times Sin\ 16}\right)$

= 28.8°

(v) For rake angle ($\propto$) = 20°

t_c = 0.218 mm, t = 0.127 m.m.

$x_5 = \frac{0.127}{0.218} = 0.582$

$\varphi = \tan^{-1}\left(\frac{0.582 \times Cos\ 20}{1-0.582 \times Sin\ 20}\right)$

= 31.18°

- Avg. speed of grinding wheel
- Effect of abrasive grain structure on accuracy & surface finish (fine, coarse)
- Benefit against conventional machining (cab be used for hard material)
- Line diagram & use of universal tool cutter griinder

Machine Tool Lab
Experiment – 1

Object:- Explain term machinability and conduct cuttability test on a drilling machine for comparing machinability of different metals

APPARATUS:- Work pieces of Aluminium, cast iron & mild steel.

Theory:-

Metal Machining The term "machinability" is a relative measure of how easily a material can be machined when compared to 160 Brinell AISI B 1112 free machining low carbon steel. The American Iron and Steel Institute (AISI) ran turning tests of this material at 180 surface feet and compared their results for B 1112 against several other materials. If B 1112 represents a 100% rating, then materials with a rating less than this level would be decidedly more difficult tomachine, while those that exceed 100% would be easier to machine.

The machinability rating of a metal takes the normal cutting speed, surface finish and tool life attained into consideration. These factors are weighted and combined to arrive at a final machinability rating. The following chart shows a variety of materials and their specific machinability ratings:

Cast Iron

All metals that contain iron (Fe) are known as ferrous materials. The word "ferrous" is by definition, "relating to or containing iron." Ferrous materials include cast iron, pig iron, wrought iron, and low carbon and alloy steels. The extensive use of cast iron and steel workpiece materials can be attributed to the fact that iron is one of the most frequently occurring elements in nature.

When iron ore and carbon are metallurgically mixed, a wide variety of workpiece materials result with a fairly unique set of physical properties. Carbon contents are altered in cast irons and steels to provide changes in hardness, yield and tensile strengths. The physical properties of cast irons and steels can be modified by changing the amount of the iron-carbon mixtures in these materials as well as their manufacturing process.

Pig iron is created after iron ore is mixed with carbon in a series of furnaces. This material can be changed further into cast iron, steel or wrought iron depending on the Cast iron is an iron carbon mixture that is generally used to pour sand castings, as opposed to making billets or bar stock. It has excellent flow properties and therefore, when it is heated to extreme temperatures, is an ideal material for complex cast shapes and intricate molds. This material is often used for automotive engine blocks, cylinder heads, valve bodies, manifolds, heavy equipment oil pans and machine bases.

Gray Cast Iron: Gray cast iron is an extremely versatile, very machinable relatively low strength cast iron used for pipe, automotive engine blocks, farm implements and fittings. This material receives its dark gray color from the excess carbon in the form of graphite flakes, which give it its name.

White Cast Iron: White cast iron occurs when all of the carbon in the casting is combined with iron to form cementite. This is an extremely hard substance that results from the rapid cooling of the casting after it is poured. Since the carbon in this material is transformed into cementite, the resulting color of the material when chipped or fractured is a silvery white. Thus the name white cast iron. However, white cast iron has almost no ductility, andtherefore when it is subjected to any type of bending or twisting loads, it fractures. The hard brittle white cast iron surface is desirable in those instances where a material with extreme abrasion resistance is required. Applications of this material would include plate rolls in a mill or rock crushers.

Yield strength is measured by pulling a test specimen as shown.

Malleable Cast Iron: When white cast iron castings are annealed (softened by heating to a controlled temperature for a specific length of time), malleable iron castings are formed. Malleable iron castings result when hard, brittle cementite in white iron castings is transformed into tempered carbon or graphite in the form of rounded nodules or aggregate. The resulting material is a strong, ductile, tough and very machinable product that is used on a broad scope of applications.

Nodular Cast Iron: Nodular or "ductile" iron is used to manufacture a wide range of automotive engine components including cam shafts, crank shafts, bearing caps and resistant.

Steel:-

Steel materials are comprised mainly of iron and carbon, often with a modest mixture of alloying elements. The biggest difference between cast iron materials and steel is the carbon content. Cast iron materials are compositions of iron and carbon, with a minimum of 1.7percent carbon to 4.5 percent carbon. Steel has a typical carbon content of .05 percent to 1.5percent.

The commercial production of a significant number of steel grades is further evidence of the demand for this versatile material. Very soft steels are used in drawing applications for automobile fenders, hoods and oil pans, while premium grade high strength steels are used for cutting tools. Steels are often selected for their electrical properties or resistance to corrosion. In other applications, non-magnetic steels are selected for wrist watches and mines weepers.

Plain Carbon Steel: This category of steels includes those materials that are a combination of iron and carbon with no alloying elements. As the carbon content in these materials is increased, the ductility (ability to stretch or elongate without breaking) of the material is reduced. Plain carbon steels are numbered in a four-digit code according to the AISI or SAE system (i.e. 10XX). The last two digits of the code indicate the carbon content of the materialin hundredths of a percentage point. For example, a 10 18 steel has a 0.18-percent carbon content.

Alloy Steels: Plain carbon steels are made up primarily of iron and carbon, while alloy steels include these same elements with many other elemental additions. The purpose of alloying steel is either to enhance the material's physical properties or its ultimate manufacturability.The physical property enhancements include improved toughness, tensile strength, hardenability, (the relative ease with which a higher hardness level can be attained), ductility and wear resistance. The use of alloying elements can alter the final grain size of a heat-treated steel, which often results in a lower machinability rating of the final product.

Tool Steels: This group of high strength steels is often used in the manufacture of cutting tools for metals, wood and other work-piece materials. In addition, these high-strength materials are used as die and punch materials due to their extreme hardness and wear resistance after heat treatment. The key to achieving the hardness, strength and wear-resistance desired for any tool steel is normally through careful heat treatment. These materials are available in a wide variety of grades with a substantial number of chemical compositions designed to satisfy specific as well as general application criteria.

Stainless Steels: As the name implies, this group of materials is designed to resist oxidation and other forms of corrosion, in addition to heat in some instances. These materials tend to have significantly greater corrosion resistance than their plain or alloy steel counterparts due to the substantial additions of chromium as an alloying element. Stainless steels are used extensively in the food processing, chemical and petroleum industries to transfer corrosive liquids between processing and storage facilities. Stainless steels can be cold formed, forged, machined, welded or extruded. This group of materials can attain relatively high strength levels when compared to plain carbon and alloy steels. Stainless steels are available in up to150 different chemical compositions. The wide selection of these materials is designed to satisfy the broad range of physical properties required by potential customers and industries. Stainless steels fall into four distinct metallurgical categories. These categories include :austenitic, ferritic, martensitic, and precipitation hardening.

Nonferrous Metals and Alloys Nonferrous metals and alloys cover a wide range of materials from the more common metals such as aluminum, copper, and magnesium, to high-strength high-temperature alloys such as tungsten, tantalum and molybdenum. Although more expensive than ferrous metals, nonferrous metals and alloys have important applications because of their numerous properties, such as corrosion resistance, high thermal and electrical conductivity, low density, and ease of fabrication. Tools. machinability are discussed below:

Tool Life: Metals that can be cut without rapid tool wear are generally thought of as being quite machinable, and vice versa. A work-piece material with many small hard inclusions may appear to have the same mechanical properties as a less abrasive metal. It may require no greater power consumption during cutting. Yet, the machinability of this material would be lower because its abrasive properties are responsible for rapid wear on the tool, resulting in higher machining costs.

One problem arising from the use of tool life as a machinability index is its sensitivity to the other machining variables. Of particular importance is the effect of tool material. Machinability ratings based on tool life cannot be compared if a high-speed steel tool is used in one case and a sintered carbide tool in another. The superior life of the carbide tool would cause the machinability of the metal cut with the steel tool to appear unfavorable. Even if identical types of tool materials are used in evaluating the work-piece materials, meaningless ratings may still result. For

example, cast iron cutting grades of carbide will not hold up when cutting steel because of excessive cratering, and steel cutting grades of carbide are not hard enough to give sufficient abrasion resistance when cutting cast iron.
Tool life may be defined as the period of time that the cutting tool performs efficiently. Many variables such as material to be machined, cutting tool material, cutting tool geometry, machine condition, cutting tool clamping, cutting speed, feed, and depth of cut, make cutting tool life determination very difficult.
The first comprehensive tool life data were reported by EW. Taylor in 1907, and his work has been the basis for later studies. Taylor showed that the relationship between cutting speed and tool life can be expressed empirically by: VT = C where: V = cutting speed, in feet per minute T = tool life, in minutes C= a constant depending on work material, and other machine variables.
This equation predicts that when plotted on log-log scales, there is a linear relationship between tool life and cutting speed. The exponent n has values ranging from 0.125 for high-speed steel (HSS) tools, to 0.70 for ceramic tools.

Tool Forces and Power Consumption: The use of tool forces or power consumption as a criterion of machinability of the work-piece material comes about for two reasons. First, the concept of machinability as the ease with which a metal is cut implies that a metal through which a tool is easily pushed should have a good machinability rating. Second, the more practical concept of machinability in terms of minimum cost per part machined, relates to forces and power consumption, and the overhead cost of a machine of proper capacity.
When using tool forces as a machinability rating, either the cutting force or the thrust force (feeding force) may be used. The cutting force is the more popular of the two since it is the force that pushes the tool through the work-piece and determines the power consumed. Although machinability ratings could be listed according to the cutting forces under a set of standard machining conditions, the data are usually presented in terms of specific energy. Work-piece materials having a high specific energy of metal removal are said to be less machinable than those with a lower specific energy.
The use of net power consumption during machining as an index of the machinability of the work-piece is similar to the use of cutting force. Again, the data are most useful in terms of specific energy. One advantage of using specific energy of metal removal as an indication of machinability, is that it is mainly a property of the work-piece material itself and is quite insensitive to tool material. By contrast, tool life is strongly dependent on tool material.
The metal removal factor is the reciprocal of the specific energy and can be used directly as a machinability rating if forces or power consumption are used to define machinability. That is, metals with a high metal removal factor could be said to have high machinability.

Surface Finish: The quality of the surface finish left on the work-piece during a cutting operation is sometimes useful in determining the machinability rating of a metal. The fundamental reason for surface roughness is the formation and sloughing off of parts of the built-up edge on the tool. Soft, ductile materials tend to form a built-up edge rather easily. Stainless steels, gas turbine alloy and other metals with high strain hardening ability also tend to machine with built-up edges. Materials which machine with high shear zone angles tend to minimize built-up edge

effects. These include the aluminum alloys, cold worked steels, free-machining steels, brass and titanium alloys. If surface finish alone were the chosen index of machinability, these latter metals would rate higher than those in the first group.
In many cases, surface finish is a meaningless criterion of work-piece machinability. In roughing cuts, for example, no attention to finish is required. In many finishing cuts, the conditions producing the desired dimension on the part will inherently provide a good finish within the engineering specification.
Machinability figures based on surface finish measurements do not always agree with figures obtained by force or tool life determinations. Stainless steels would have a low rating by any of these standards, while aluminum alloys would be rated high. Titanium alloys would have a high rating by finish measurements, low by tool life tests, and intermediate by force readings.
The machinability rating of various materials by surface finish are easily determined. Surface finish readings are taken with an appropriate instrument after standard work-pieces of various materials are machined under controlled cutting conditions. The machinability rating varies inversely with the instrument reading. A low reading means good finish, and thus high machinability. Relative ratings may be obtained by comparing the observed value of surface finish with that of a material chosen as the reference.

Machinability depends on:-

1. The type of work piece i.e. its composition
2. Type of tool material.
3. Size & shape of tool.
4. Type of machining operation.
5. Size, shape & velocity of cut.
6. Type & quality of machine used.
7. Quality of lubricant used during machining operation.
8. Coefficient of friction between chip & tool.
9. Shearing strength of wok piece material.

Machinability can also be considered as the fase with which a given material can be machined & affected by m/c variables like cutting speed, feed & depth of cut, the material, cutting fluid, rigidity of m/c tool, shape & size of work. For a given set of machine conditions machining is affected by properties of work material. Like hardness, tensile properties, chemical composition micro structure etc.

The factors which come into play while evaluation of the amenability of any metal are:-

1. Tool life between grinds.
2. Value of cutting forces.
3. Surface finish.
4. Size of chips.
5. Metal removal rate.
6. Temperature of cutting.
7. Rate of cutting under standard force

8. Uniformity in dimensional occurs of successive part.

The main factors to be chosen for judging machinability depends on the type of operation & production requirement. The life is the most important parameter for assuring machinability. Since tool life is a direct function of cutting speed a better machinable metal is one which permits higher cutting speed for a given tool life.

MACHINABILITY INDEX: -

Machinability is a relative term. The rated machinability of two or more metals being compared may way for different processes of machining such as heavy or light turning milling drilling etc.

The machinability of different metals to be machined may be comparable by using the machinability index of each. This is Machinability Index.

$$= \frac{Cutting\ speed\ of\ material\ for\ 20\ min.tool\ life}{Cutting\ speed\ of\ standard\ steel\ for\ 20\ min.\ tool\ life}$$

The standard steel is a free cutting steel, which machinability easily & machinability index of which is arbitrarily fixed at 100%. This steel has carbon content of 0.13% max. min. 0.06-1.1% & 5 0.08-0.03%.

1. *Standard tests*

1.1. *ISO (ASME) test*

Although often referred to as machinability standards, the international standard ISO 3685 “Tool-Life Testing with Single-Point Turning Tools” and its analog ANSI/ASME “Tool-life Testing With Single-Point Turning Tools (B94.55M-1985) can hardly be considered as directly related to machinability because these standards present the rather obsolete methodology for tool-life testing where one parameter is changed at a time [AST 04]. Both standards consider the rake and flank tool wear types and patterns as they are well described in the literature on metal cutting [AST 06, AST 04, AST 08a, SHA 84].

Standard tool life testing and representation includes Taylor’s tool-life formula:
$vT^n = C_T$

where *v* is the cutting speed in meters per minute, *T* is the tool life in minutes, *CT* is a constant into which all cutting conditions affecting tool life must be absorbed. Although Taylor’s tool-life formula is still in wide use today and is at the very core of many studies on metal cutting including at the level of national and international standards, one should remember that it was introduced in 1907 as a generalization of 26 years of experimental studies conducted in the 19th Century using the work and tool materials and experimental techniques available at that time. Since then, each of these three components has undergone dramatic changes. Unfortunately, the validity of the formula has

never been verified for these new conditions. So far, nobody has proven that it is still valid for any cutting tool materials other than carbon steels and high-speed steels, for cutting speeds higher
than 25 m/min.

Figure 2 shows the experimental procedure of determining Taylor's formula coefficients according to standard ANSI/*ASME* B94.55M-1985 for three cutting speeds *v1* > *v2* > v3 [AST 08a]. A simple analysis of Taylor's tool-life formula shows that it actually correlates cutting temperature with tool life as the cutting speed solely determines the cutting temperature. As can be seen, this formula states that the higher the cutting speed (temperature), the lower the tool life, which is in direct contradiction with well-known experimental studies and the practice of metal cutting [AST 08a]. Leading tool manufacturers clearly indicate the favorable range of cutting speeds (temperatures) for their tool materials. Deviation from the recommended speed (temperature) for a given tool material on either side lowers tool life. This, however, does not follow from Taylor's tool-life formula.

Tool life, considered according to the standards as the operating time until the selected tool failure criterion is reached, does not reflect the cutting regime and thus does not reflect the real amount of work material removed by the tool during the time over which the measured flank wear is achieved. In this sense, this tool life does not have much meaning. Moreover, this tool life is particular and thus, in general, is not suitable for the optimization of machining operations, the comparison of various cutting regimes, the assessment of various tool materials and so on. For example, it is not possible to compare two different tool materials if two different cutting speeds (suitable respectively for each one in particular) were used in the tests.

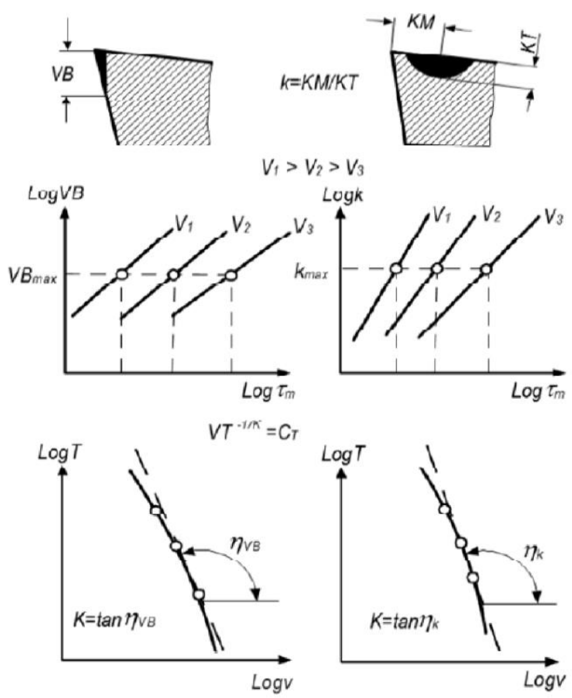


Figure 2. *Experimental procedure of determining of Taylor's formula coefficients according to standard ANSI/ASME B94.55M-1985*

Another significant drawback of this methodology for machinability assessment is that tool life is considered as the only parameter of machinability leaving other essential parameters, for example surface integrity, accuracy of machined parts, chip control, etc., out of consideration. In modern manufacturing where the machining quality and process efficiency are of prime concern and where advanced machines with spindles and controllers are capable of measuring force factors and in-process machining quality, tool wear is no longer considered to be the prime criterion of machinability. Instead, a certain combination of quality parameters such as, for example, diametric accuracy and surface roughness, are of prime concern. So machinability is "silently" considered to be the ability of a given machining operation to achieve the pre-set quality requirements for a given work material while keeping a pre-defined level of process efficiency.

1.2. *ASTM test*

Standard ASTM E618-07 "Standard Test Method for Evaluating Machining Performance of Ferrous Metals Using an Automatic Screw/Bar Machine" was written to fill a requirement for a standard test for determining the machinability of ferrous metals using automatic screw/bar machines. Machinability is considered to be a suitability of a particular work material for producing parts of the standard design in such machines to a uniform level of quality with respect to surface roughness and size variation. The standard intends to simulate mass production conditions in a controlled environment using a single- or multi-spindle automatic screw machine. The essence of the standard is to compare production rate in manufacturing the standard part shown in Figure 1.3. The part is designed to make use of the three most common screw machining operations: rough turning, finish turning and twist drilling. The part is machined from bar stock of 1" (25.4 mm) dia. The diametric tolerances and surface roughness are the control parameters.

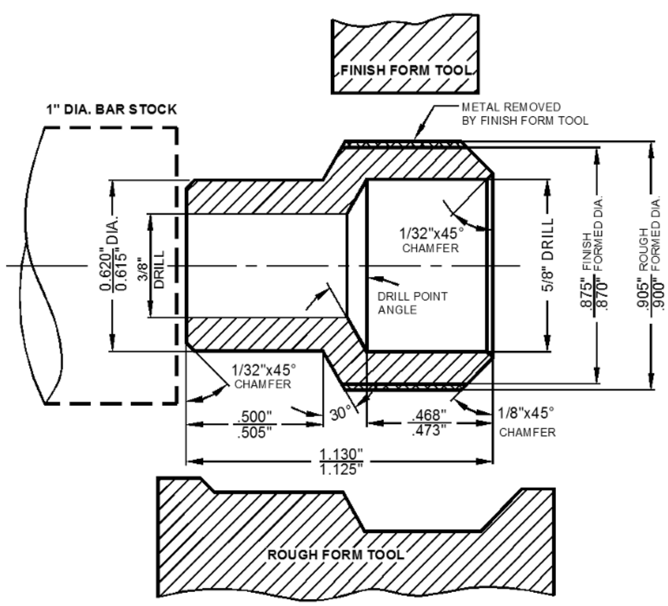


Figure 3. *Automatic screw machine part required for ASTM E618*

The machinability of a material is measured by the maximum production rate at which test pieces can be produced to specified surface roughness and size limits for specific periods of time and by the cutting speed and tool-feed rate to attain this production rate. The tool geometry and the tool materials (HSS M2 for the form tools and M7 for the drills) are fixed by the standard. The machining sequence is specified by the standard and is based on normal industrial practice. Some recommendations for the calibration of a screw machine to be used for testing are also given.

The standard points out that "the machining performance (or, as it sometimes called, the "machinability") of material cannot be regarded solely as a property characteristic of that material. The principal indexes of machining performance, namely, production rates, cutting speeds, and tool-feed rates, are generally affected by many other factors, such as the tool material, the surface roughness, and dimensional limits demanded of the product, the coolant and its properties, and the configuration of the part. These latter factors are quite independent of the work material and yet all affect its machining performance criteria."

The standard admits that testing according to this standard is neither simple nor inexpensive. Substantial quantities of the test material are required, varying from a few hundred kilograms to a few thousand. A significant amount of time is required to compare different work materials including finding the proper regimes, the machining itself, and part quality evaluation.

The standard is the first, and so far the only, attempt to evaluate machinability instead of other output characteristics of the metal cutting process. It provides a detailed testing methodology that concerns various aspects of the metal cutting system including the determination of machine capability, test piece design, cutting tool design and geometry, etc. The standard has a number of obvious shortcomings:

- It uses obsolete tool designs and tool material that are hardly used in today's manufacturing practice. The eight-hour tool life required by the standard is not efficient for modern machine tools.
- The most uncertain place in the test methodology suggested by the standard is the selection of the cutting speed and cutting feed (tool-feed rate) to assure an eighthour tool life for all three tools (the rough and finish form tools and twist drill). The standard recommends such a selection on the basis of experience or general guidelines for a ferrous metal of similar composition and conditions. However, if one knows the cutting speed and cutting feed to assure an eight-hour tool life, the reason for conducting the test at the production rate required by the standard at the tests' outcome can simply be calculated. On the other hand, if these parameters are unknown, their determination by the trial-and-error methods may take virtually forever, including tons of wasted work material and many machine and man hours.

1.3. *American Iron and Steel Institute (AISI) test*

The AISI Bar Machinability Subcommittee was formed in 1991 and was composed of representatives from automotive OEMs, academia and the steel industry. It aimed to develop information needed by the machining industry for material selection, process development and for improving the understanding of the factors that influence the machinability of steel. To accomplish this task, more than 30 industrially significant steel grades and their variants were evaluated in the ensuing years.

The test materials were produced by eight different steel companies using various melting and casting practices. Material properties and microstructures were characterized and the machinability of each steel variant was evaluated by at least two different machinability testing laboratories. A study of the machinability of more than 30 industrially significant carbon, alloy, resulfurized and microalloyed steel grades using carbide tools in a standardized single point turning test was conducted. It was found that machining data generated with high-speed steel tooling could not be directly extrapolated to applications involving carbide tooling. The plain carbon and alloy steels were found to have a *v30*-tool life that correlated well with their Ito-Bessyo Carbon Equivalent when fitted to a 3rd order polynomial. It was also found that the machinability of 1200 series, 1100 series, microalloyed and leaded steels followed the same relationship. The values of the cutting speed corresponding to a 30-min tool life, *v30*, were generated in this study and suggested to be used for guidance in selecting machining parameters for the steel grades tested. The *v30* tool life of other steel grades can be approximated by calculating their Ito- Bessyo Carbon Equivalent. It is suggested that the database thus generated can be used by the machining industry to compare the relative machinability of various steel grades and their properties to make more informed decisions about the application of materials.

In establishing a machinability test standard, a number of factors were considered based on the discussions of the AISI Machinability Roundtable participants. First, the test procedure must not be so complex that it discourages its use. The ISO 3685 standard is quite complete, essentially covering all aspects of single point turning. However, it was the general consensus of the committee that only those features of the ISO 3685 relating to turning tests conducted with carbide tooling be addressed in the current standard. The standard test must be easy to conduct, and the cutting conditions well defined and clearly specified. A second concern was one of the reliability and transportability of standard test data. This was addressed in a round-robin series of turning tests conducted with SAE1141, SAE1541 and SAE4140 steels. The preliminary tests established the reproducibility of the testing based on the proposed standard. To ensure this level of reproducibility continued between the different testing laboratories, a standard baseline material (SAE1045) was selected as a reference. All participating laboratories conducted the standard turning test on this material.

In the author's opinion, the result of this enormous effort is rather humble. The Machinability Estimator was developed for carbon and alloy steels using uncoated Machinability: Existing and Advanced Concepts 15 carbide tools and is available online in the form of an Excel table2. One simply needs to input the chemical composition of a steel, i.e. the content of carbon, silicon, manganese, sulfur, nickel, chromium, molybdenum, copper and vanadium. The estimator will return a value of the Ito-Bessyo carbon equivalent and the estimated cutting speed *v30*. No other essentials for machinability mechanical properties, such as hardness, strength, grain size, etc. (which vary significantly with heat treatment of a steel), no tool material nor its geometry, no MWF parameters nor other essentials of a particular machining operation are accounted for.

1.4. *Two kinds of machinability*

First, the machinability of work material which should be considered as an inherent property of the work material related to its physico-mechanical properties. Second, the process machinability of material that relates to the machinability of the material in a specific machining operation. Although these two notions are closely related, they are not nearly the same. The first should be considered as the ultimate goal in the optimization of the metal cutting process, while the second relates to the reduction of the first machinability due to the real-world process efficiency.

The two notions of machinability introduced can be used for:

- the development of new materials with enhanced machinability without compromising their service properties;
- assessments of the machinability of existing materials and to point out possible the direction of machinability improvements;
- assessments and optimization of the metal cutting process and operation efficiency through the concept of process machinability and physical efficiency of the cutting system.

OBSERVAION TABLES

Sr. No.	**RPM**	**Weight (Kg.)**	**TIME TO DRILL A THROUGH HOLE**	
			Aluminium workpiece (sec.) CAST IRON work piece (SEC)	

Sr. No.	**RPM**	**Weight**	**Time(S) MILDSTEEL**

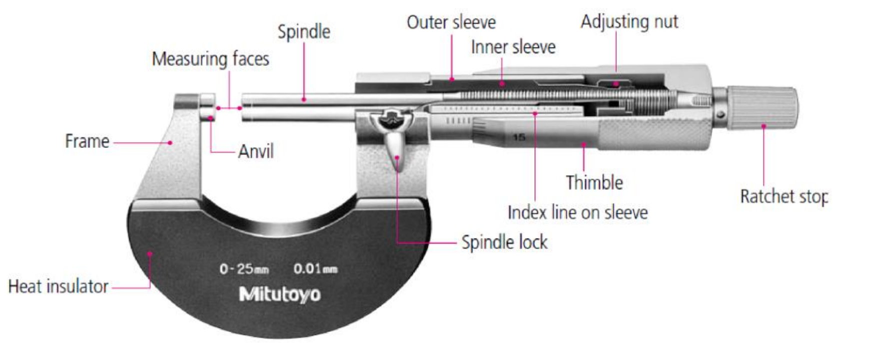
Result :-

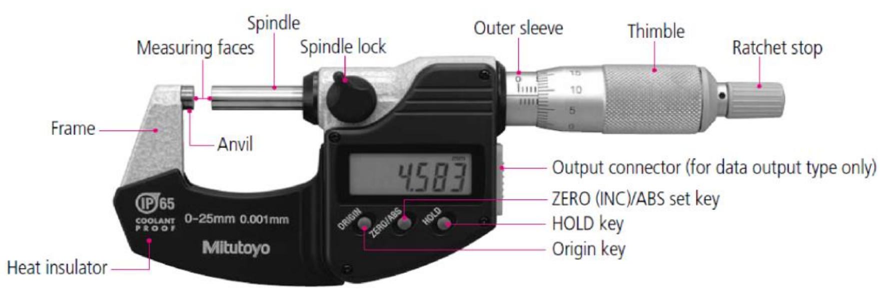
Experiment – 2

Object:- To study the tool makers microscope and checking the alignments of screw thread and cutting tool (five elements of workpiece are checked, outer dia, root dia, pitch, depth, thread angel) (measuring angles end cutting edge angle, side cutting edge angle, lip angle, top back rake angle, clearance angle)

Apparatus:- Tool makers microscope, 6′ bolt, cutting tool.

Theory:- The tool maker microscope is a versatile instrument that measure dimensions by ooptial means without application of pressure, that's why its usually used to measure small & sort material past.





It is suitable for following fields for applications:-

1.) Length (linear) measurement of tools, threading tools, hobs, punches, gauge etc.
2.) Angular measurement ie. profile, major and minor dia, height of lead screw, thread angle, profile position w.r.t. thread axis, shape of thread (rounding, flattening)
3.) Comparison b/w contours and draw profile patters.
4.) Drawing of projected profile.
5.) Measurement of distance btw. point.

1. Function :

The work-piece to be checked is arranged in the path or ray or lighting equipment it produes a shadow image plane of the objective is viewed with eyepiece having either a suitable mark for aiming at the test point of object or in case of the after occurring profile eg. thread wounding standard line pattern for comparing with the shadow image of test object after varying non-standardized profile con be checked if the shadow image of test projected to groundless screw equipped with drawing on transparent boil enlarged in the corresponding scale the test object is shifted or turndown a measuring table by means of micrometer screw permitting measuring in addition to comparison of shape.

In addition to this method, the scrolled shadow image method measuring operation are possible by use of axial section method, which can be recommended especially for thread measuring.

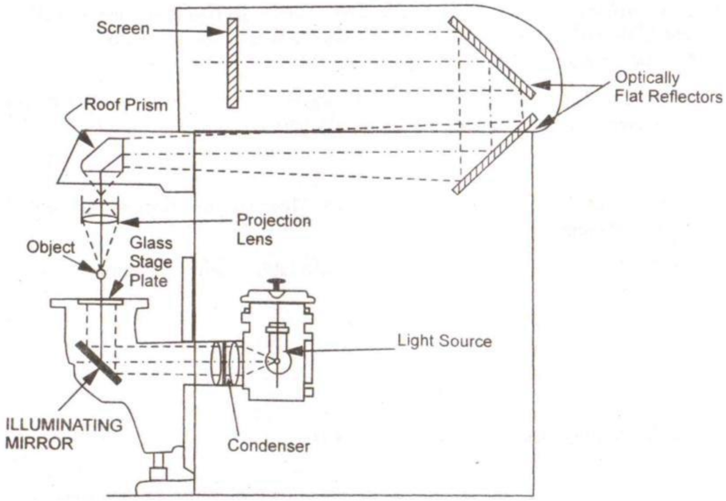


Figure 1: Optical System of Bausch and Lomb Projector

2. Data:

Scale value	**Millimeter design**	**Inch design**
• Micrometer screws	0.01 m.m.	0.0001′
• Angular graduation in selectable eyepiece	10′	10′
Measuring range		
Slide for x – direction		
• With micrometer screw only	0.0025 m.m.	0.001′
• With micrometer screw and setting gauge	0.0015 m.m.	0.006′
Slide for y – direction		
• With micrometer screw only	0.0025 m.m.	0.001′
• With micrometer & setting gauge	0.0052 m.m.	0.002′
• Angular graduation in the goinometer eyepiece	0.0036°	0.0030°
• Angular graduation in rotary table	0.0036°	0.0036°
• Swivel range of the microscope carrier	$\simeq \pm 12°$	$\simeq \pm 12°$
• Diameter of rotary table	280 m.m.	11′
• Overhang of microscope arm	$\simeq$ 167	$\simeq$ 6.58′
• Max^{m} mounting length in the centre stand		
(i) For object upto diameter 3.9 m.m. (1.53′)	$\simeq$ 315 mm	$\simeq$ 12.41′
(ii) For object upto diameter 40–85 m.m.	$\simeq$ 235 mm	$\simeq$ 9.25′

Instrument diamension:

- width $\simeq$ 510 m.m.
- length $\simeq$ 900 m.m.

- height (without projection drive) ≃ 550 m.m.
- weight or complete equipment ≃ 50 k.g.

Selectable Eyepiece:

- Virtual graduation interval size : 1.1 m.m.
- Range of angle measuring with eyepiece : ± 7°
  C2, H2, So2, N2, O, P2, Q, R W3 8 7
- Graduation value of angular graduation : 10′
- Eyepiece magnification : ×10
- Entire magnification according to the objective used : ×10, ×15, ×30, ×50
- Viewed section according to the objective used : 21, 14, 7, 4 mm

Goniometer Eyepiece:

- Scale value of course graduation : 1′
- Scale value of sensitive qraduation : 1′
- Eyepiece magnification : ×10
- Magnification of microscope for Graduated circle reading : ×40

Measuring lever addition:

- Swivel range of feller : ±30
- Max^{m} depth of immersion : 15 mm
- Measuring force : ≃ 10 N
- Max^{m} reversible discrepancy range : 1.3 μm

3. <u>Measuring error of the device :</u>

Tool makers microscope has un-accuracies exceeding the value mentioned in the following. These error have been calculated on the basis of disadvantage condition and represal maxima to be expected:

Shadow image method (length measurement)

in x – dir : $\pm\left(3.5 + 0.7k + \frac{L}{46} + \frac{H \times L}{1500}\right) \mu\text{m}$

in y – dir : $\pm\left(3.5 + 0.7k + \frac{L}{50} + \frac{H \times L}{2000}\right) \mu\text{m}$

Angle measurement of flat component with

Goniometer eyepiece : $\pm\left(1.0 + \frac{1.5}{F}\right)$

Observation table:

Piter = 14.577 – 11.979

= 2.508 m.m.

Outer dia = 19.829 – 4.973

= 14.850 m.m.

. root dia = 17.659 – 2.771

= 14.888

Depth = 19.829 – 17.659

= 2.17

Angle = 302°.1′ – 240°18′

= 55°49′

Single point cutting tool:

- Side cutting edge angle : 93.35° - 80.15° = 12.2°
- End cutting edge angle : 90 – (156.5 – 92.65) = 23.15°
- Nose angle : 151.55° - 81.15 = 70°.40′
- Top rake angle : 94°.80′ - 77°.28′ = 17°.52′
- End relief angle : 90° - (96°.15′ - 30°.65′) = 24°. 5′

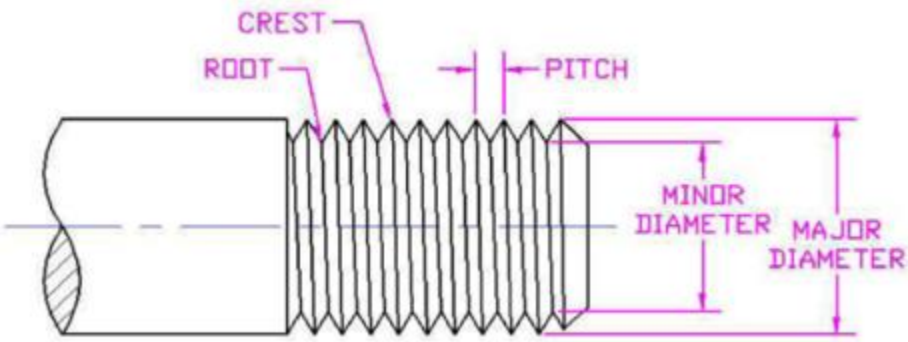


Fig. 2 External Threads

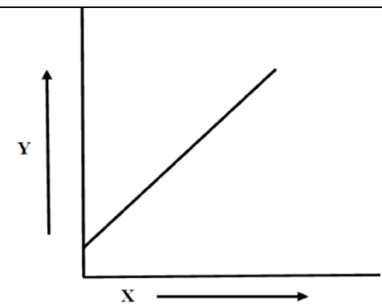
Experiment – 3

Object: *To study the pneumatic comparator, calibration and find out the inclination of a given piece.*

Equipment: *Solex type pneumatic comparator and compressor*

Calibration the Pneumatic Comparator: Calibration of pneumatic comparator is down because it helps and guides in taking the reading during regular use and we are also able to understand the taking the initial reading (range) of the system.

Table 5.1: Reading for Calibrations

S. No.	Dial Gauge Reading (X)	Manometer Reading (Y)	Graph Characteristic curve S v/s Y
1.	0.0		
2.	0.1		
3.	0.2		
4.	0.3		
5.	0.4		
6.	0.5		
7.	0.6		
8.	0.7		

Method of Calibration: By starting the compressor and supplying compressed air at constant pressure. This air passes through the controlling and measuring orifices. We start taking readings from the manometer by using dial gauge for the required gap between the measuring orifice and specimen. After knowing the initial reading from where the back pressure start and level of water in manometer tube falling down. We take the readings by suing dial gauge in 0.1 mm reducing gap between measuring orifice and specimen till upto the surface of the measuring orifice. The table & graph are plotted as required.

5.4 Pneumatic comparator:

The pneumatic back pressure sensor the object and provides a pressure signal. The intensity of back pressure signal from the sensor depends upon the relative position of the object from the sensor provided supply pressure to the sensor is maintained constant. Hence, for a slight change in the distance of the object of the sensor ranges in microns gives a considerable change in back pressure and thus sensor becomes indispensable in the field of repetitive gauging to close tolerance. The instrument make use of a constant supply of air at controlled pressure. The gauging element is based upon the principle of allowing elative small quantity of air to escape between the outside or inside of the gauging element and the work. The velocity or pressure variation caused by the escaped air is then measured on a sensitive amplifying device. In most application the pressure variation are measured and the pneumatic comparator unit is calibrated against master gauge so that the measurement obtained with the gauging head represents deviation from the master gauge dimension thus the pneumatic gauge is basically compared at measuring difference and not basic or standard dimensions.

5.5 Operating principle:

Back pressure sensor is most commonly used position sensor. The principle of pneumatic nozzle and tapper device is utilized. The sensor consists of a constant pressure supply separated from back pressure chamber by a fixed restriction with form or control orifice. The sensor is shown figure when the distance of the object changes with nozzle =, back pressure that is the pressure reflected on upstream side of a system due to downstream operation, becomes the function of displacement of the object, provided supply pressure remains constant. The backpressure characteristics i.e., graph between back pressure/supply pressure v/s non dimensional displacements is shown in figure. It is seen that a portion of curve which is approximately leaner can be used for a sensing system. For digital system application the extreme point of the characteristics is only points of interest so that a pressure is attained above and below the switching pressure of device has wider application in air gauging to closed tolerance and in proximity sensing. It is claimed that by using backpressure sensor under controlled condition a precession gauging of a few microns can be obtained.

5.6 Study of solex pneumatic comparator

The principle of this type of measuring instrument depends upon the air flow between the gauging head and hole in which it is inserted. Main parts of the solex pneumatic comparator are as below:

1. Threaded spindle
2. Knurled nuts
3. Socket
4. Measuring orifice
5. Controlled orifice
6. Water tank chamber
7. Air chamber
8. Wooden box
9. Prime mover (compressor)

Referring to figure air at a under constant pressure obtained **A** dip tube controlling device passes through a control jet **B** which meters the amount of air delivered to the gauge. At **C** there is flexible air tight tube between **B** and the gauging head jet **G.** A manometer or air pressure gauge is connected to the branch pipe at **D**. The intermediate chamber between **B and G** is indicated as **F**, the work to be measured at **I** and the instrument stand at **E.**

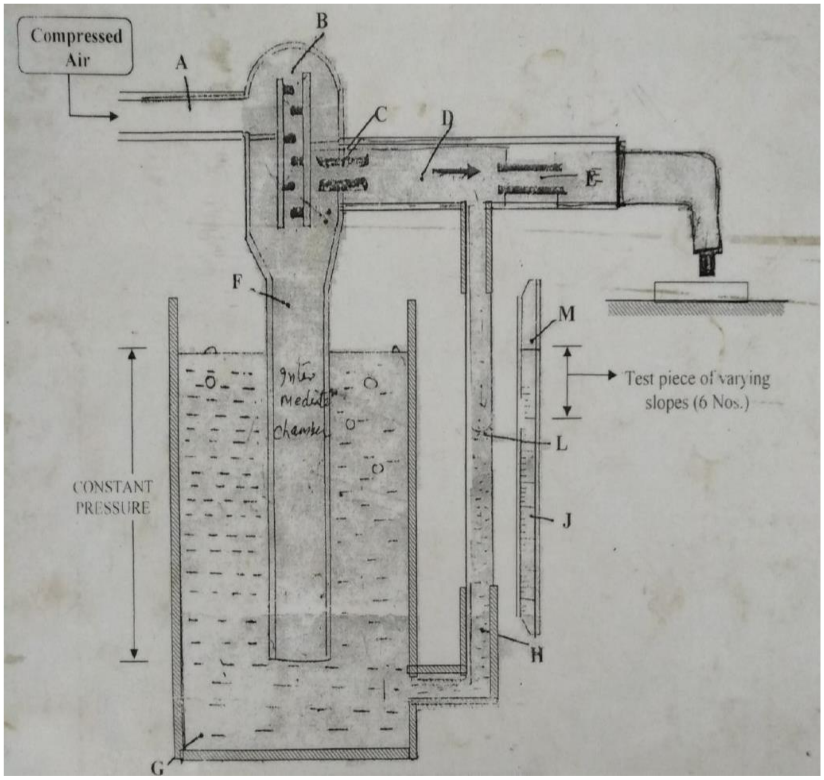


Fig 1: Air pressure regulator

Air under pressure from a main supply is set at a constant pressure and then flows through the control jet **B** to the orifice known as the gauging head at **G.** According to the amount of air escaping through the latter jet a pressure is built up to definite value between **B** and **G,** so that the flow through them are exactly balanced and are independent of outside conditions. The pressure depends only upon the relative value of the orifices and the control jet B is constant, the pressure than depends only upon the second orifice E i.e. the clearance between the gauging head and the work which is the example shown in figure is a flat surface parallel to the lower suface of the gauging head. The manometer measures this pressure which is then taken as a measure of the clearance between the gauging head and the work. The air pressure regulator works on the dip tube principal shown at **A** is the connection to main air supply at **B** certain restrictions jets, to limit the maximum amount of air delivered by air line the control jet is shown at **C** the intermediate chamber at flexible connection to the gauging head **G** dip type at F water container at **G** manometer at J and glass tube of manometer at **H** with water level at **I.**

In use air from the main supply at a restricted in amount by B passes down the dip tube F and just bubbles out, this maintaining constant pressure in dip tube. The actual value of the is pressure is that of the head of water above the bottom and of the dip tube. When the gauging head in operation the pressure in the intermediate chamber will be less than that in the dip tube its value being shown B, The vertical distance between the water level M and lower part scale L of the water in the manometer tube. The scale of the letter is graduated according to the dimensions of part to be measured and each gauge has its own particular scale.

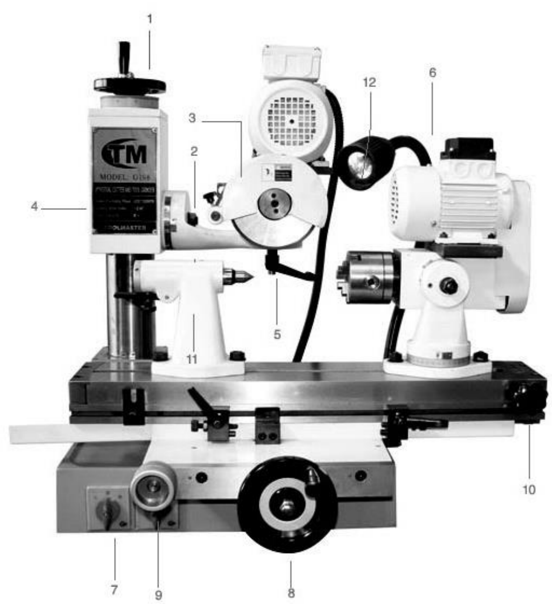
Experiment – 4

Object:- Study of universal tool cutter grinder

Theory:-

Grinding machine is basically a machine tool. The distinguishing feature of grinding machine is that, it is using a grinding wheel, which is rotor in nature and operating at very high speed. This is primarily used for achieving high accuracy and high finish which can't be achieved by normal machining. This is also one process which can be used on the work material which are hardened on which normal conventional machining can't be performed.

One of them is tool and cutter grinding. Tool & cutter grinding is necessary for two reasons. One to sharpen the cutter or the tool which is already blunt. Another to produce these tools with proper geometry, after the heat treatment hardening this needs proper geometry on the cutting point or the tip of the tool and that needs proper grinding and this is used for actually the production of the wheel and in this case we have least two types of tool and cutter grinder. One is the pedestal or bends type grinder and another is universal tool and cutter grinder.



I. Elevating hand-wheel

2. Wheel head tilt clamp.

3. Super-Precision motor spindle assembly.

4. Vertical locking screw.

5. Wheel head setting lever.

6. Motorized work head

7. Forward and reverse switches.

8. Cross feed hand wheel

9. Longitudinal hand wheel

10. Table Taper adjustment
11. Tailstock
12. Work light

Universal tool cutter grinder: These are the universal tool cutter grinder where we can use a three-dimensional vice. It is a fixture which can be swiveled in space in three orientations. It has three axes; the particular geometry of cutter can be set and here no particular scale is necessary. The person who understands the basic geometry of grinding or cutting tools or the cutter can easily set a vice in proper angle for the necessary grinding action.